THE PROPERTY OF BEING A HENSELIAN PAIR IS PRESERVED BY LIMITS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Limits are built from products and equalizers. Categories, Lemma \ref{categories-lemma-limits-products-equalizers}, cited at proof lines 2852-2853. Reduce the general statement to the product and equalizer cases.
- A product of henselian pairs is henselian. Lemma \ref{lemma-product-henselian-pairs}, cited at proof lines 2854-2855. Settles the product case.
- Gabber's criterion for a henselian pair. Lemma \ref{lemma-characterize-henselian-pair}, cited at proof lines 2860-2861. Provides the property of the polynomial that has to be verified for the limit pair.
- Elements of one plus an ideal contained in the Jacobson radical are units. Algebra, Lemma \ref{algebra-lemma-contained-in-radical}, cited at proof lines 2866-2867. Shows the limit ideal lies in the Jacobson radical once units are checked.

Source reference. Original lemma 0EM6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18296–18296. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 002N: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 104–104. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0ATD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13611–13611. Navigation pointer to a statement the

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Source reference. Original lemma 09XI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12531–12531. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0AME: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13402–13402. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *The property of being Henselian is preserved under limits of pairs. More precisely, let J be a preordered set and let (A_j, I_j) be an inverse system of henselian pairs over J. Then $A = \lim A_j$ equipped with the ideal $I = \lim I_j$ is a henselian pair (A, I).*

Proof. By Categories, Lemma 002N, we only need to consider products and equalizers. For products, the claim follows from Lemma 0ATD. Thus, consider an equalizer diagram

$$(A, I) \longrightarrow (A', I') \underset{\psi}{\overset{\varphi}{\rightrightarrows}} (A'', I'')$$

in which the pairs (A', I') and (A'', I'') are henselian. To check that the pair (A, I) is also henselian, we will use the Gabber's criterion in Lemma 09XI. Every element of $1 + I$ is a unit in A because, due to the uniqueness of the inverses of units, this may be checked in (A', I'). Thus I is contained in the Jacobson radical of A, see Algebra, Lemma 0AME. Thus, let

$$f(T) = T^{N-1}(T - 1) + a_{N-1}T^{N-1} + \cdots + a_1 T + a_0$$

be a polynomial in $A[T]$ with $a_{N-1}, \ldots, a_0 \in I$ and $N \geq 1$. The image of $f(T)$ in $A'[T]$ has a unique root $\alpha' \in 1 + I'$ and likewise for the further image in $A''[T]$. Thus, due to the uniqueness, $\varphi(\alpha') = \psi(\alpha')$, to the effect that α' defines a root of $f(T)$ in $1 + I$, as desired. □